

WEEK 9 ULTIMATE MASTER GUIDE

NPTEL: Introduction to Environmental Engineering & Science

Exam Target: April 17, 2026 | **Focus:** Wastewater Characteristics, BOD, Preliminary/Primary Treatment & Biological Treatment

SECTION 1: THE FINAL EXAM TRAP LIST

- **The Biodegradability Trap:** If BOD_u / COD is between **0.92 and 1.0**, or if BOD_5 / COD is between **0.63 and 0.68**, the waste is considered **fully biodegradable**.
 - **The Trickling Filter Trap:** A trickling filter is **NOT** a physical strainer or filter. It is a biological reactor. The rocks simply provide surface area for the microbial slime layer to grow on.
 - **The Anaerobic Trap:** A **Septic Tank** operates **anaerobically**. An **Oxidation Pond** operates on **algae-bacterial symbiosis** (aerobic/facultative).
 - **The NRC Unit Trap:** In the NRC efficiency formula, the organic loading term u MUST be in **kg/ha·m/day**. The filter volume (V) is in **hectare-meters (ha·m)**.
 - **The Combined Sewer Trap:** The combined system's critical flaw is that storm events overwhelm the treatment plant, forcing it to **bypass raw untreated sewage** directly into the river.
-

SECTION 2: THE SEWER PIPES & WATER FLOW

1. Combined vs. Separate Sewers:

- **Separate System:** One pipe for rainwater (stormwater), one pipe for toilet water (sanitary sewage).
- **Combined System:** One giant pipe for both. **Disadvantage:** During a massive storm, the treatment plant gets overwhelmed with too much water (Hydraulic Load) and has to bypass raw, untreated sewage directly into the river.

2. The Sewer Hierarchy (order matters!):

House Sewer Lateral Sewer Trunk/Main Sewer Outfall Sewer (which dumps into the river)

- **Self-Cleansing Velocity:** To prevent solids from settling and clogging the pipes, sewers must maintain a minimum velocity.
 - **Sanitary Sewers:** ~0.6 m/s.
 - **Storm Sewers:** ~0.9 m/s.

3. Types of Wastewater:

- **Domestic:** From homes.
 - **ICI:** From factories, restaurants, and hospitals (Industrial, Commercial, Institutional).
-

SECTION 3: WHAT'S IN THE WATER?

4. The Size of Solids (Slide Veto!):

Type	Size	Behaviour
Suspended	> 1 μm	Settle by gravity
Colloidal	1 μm to 10^{-3} μm	Do NOT settle; need coagulation or biological treatment
Dissolved	< 10^{-3} μm	Invisible, fully dissolved

5. Dissolved Oxygen (DO) Traps:

- **Temperature Trap:** As water gets hotter, oxygen escapes. Cold water holds more oxygen.
- **Salinity Trap:** Freshwater holds more dissolved oxygen than seawater at the exact same temperature.

6. Population Equivalent: A mathematical trick to express factory pollution in human terms. We divide the factory's total BOD by what a normal human produces in a day (**0.08 kg BOD/day/person**). This tells us how many "people" the factory equals.

SECTION 4: THE BOD CURVE

- **The Seed Correction Trap:** If the dilution water itself contains bacteria (is "seeded"), you must subtract the oxygen they breathe from the total drop.
 - **Formula:** $\text{BOD}_5 = \frac{(\text{DO}_i - \text{DO}_f) - (\text{B}_i - \text{B}_f)f}{P}$
 - *Where:* B is the DO of the seed control, and f is the ratio of seed in the sample to seed in the control.
- **BOD Exerted (y_t):** This is the amount of oxygen actually *consumed* by time t.

- **Formula:** $y_t = L_0(1 - 10^{-kt})$.
- *Note:* If the base is 'e', use $y_t = L_0(1 - e^{-kt})$. Always check if the question gives k in base 10 or base e.

7. The 4 BOD “Vitamins” (Slide Veto!): When setting up a BOD test, you must add 4 specific nutrients to the dilution water:

1. Calcium chloride
2. Magnesium sulphate
3. Phosphate buffer
4. Ferric chloride

- **The Seed Correction Trap:** If the dilution water itself contains bacteria (is “seeded”), you must subtract the oxygen they breathe from the total drop.
 - **Formula:** $BOD_5 = \frac{(DO_i - DO_f) - (B_i - B_f)f}{P}$
 - *Where:* B is the DO of the seed control, and f is the ratio of seed in the sample to seed in the control.
- **BOD Exerted (y_t):** This is the amount of oxygen actually *consumed* by time t .
 - **Formula:** $y_t = L_0(1 - 10^{-kt})$.
 - *Note:* If the base is 'e', use $y_t = L_0(1 - e^{-kt})$. Always check if the question gives k in base 10 or base e.

8. Temperature Correction (K_T):

- Heating the water makes bacteria hyperactive **K increases.**
- Heating does NOT change the total “food” **Ultimate BOD (L_0) remains constant.**
- **Formula:**

$$K_T = K_{20} \times (1.047)^{(T-20)}$$

9. The BOD/COD Ratio (Slide Veto!):

Ratio	Range	Meaning
BOD _u / COD	0.92 – 1.0	Fully biodegradable
BOD ₅ / COD	0.63 – 0.68	Fully biodegradable

Two stages of the BOD curve:

- **Stage 1 (CBOD):** Carbonaceous oxygen demand — heterotrophic bacteria oxidize organic carbon.
 - **Stage 2 (NBOD):** Nitrogenous oxygen demand — nitrification of ammonia.
-

SECTION 5: PRELIMINARY & PRIMARY TREATMENT

10. Preliminary Treatment: Includes three processes:

- **Screening:** Catching rags and plastics.
- **Skimming:** Scraping off floating oil and grease to protect the activated sludge process downstream.
- **Grit Chambers:** Removing heavy inorganic grit (sand).

11. The Grit Chamber & Proportional Weir:

The goal is to let heavy sand (grit) fall, but keep lighter organics (poop) floating. The water must move at exactly **0.2 – 0.3 m/s**.

- **The Fix:** A **Proportional Weir (Sutro Weir)** or **Parshall Flume** is installed at the end. Its special shape keeps velocity perfectly constant even if a massive storm surges through.
- **Parshall Flumes** are often preferred over weirs because they are **self-cleaning** and cause **less head loss**.
- **Trap:** If velocity drops too low lighter organic matter settles out along with grit septic conditions and severe odors.

12. Primary Sedimentation:

- Uses **Type II (Flocculent) Settling**.
 - Particles crash into each other, stick together, and fall faster as they sink.
-

SECTION 6: SECONDARY (BIOLOGICAL) TREATMENT

13. Trickling Filters:

It is **NOT** a physical strainer! It is a tower of rocks covered in a slimy “stomach” (microbial slime layer). Dirty water trickles over the rocks, and the slime **adsorbs** and **aerobically oxidizes** the organics.

- **Ponding:** If the slime grows too thick, it clogs the rock voids. Fix: dump **Chlorine** or **Copper Sulphate** to kill excess algae/fungi.
- **Recirculation (Slide Veto!):** Looping water back through the filter helps in:
 - Seeding the sewage with bacteria
 - Accelerating oxidation
 - Reducing odors and fly nuisance (*Psychoda*)
 - **Trap:** Recirculation **increases** the hydraulic load on the filter — it does NOT reduce it.

Trickling Filter Flow Sequence:

Influent Primary Clarifier Trickling Filter Secondary Clarifier

(Sludge is returned from the secondary clarifier back to the primary clarifier, unlike the Activated Sludge Process.)

The NRC Efficiency Formula:

$$\eta = 100 / (1 + 0.0044 \times \sqrt{u})$$

Where u is the **organic loading** in **kg/ha·m/day** and volume (V) is in **hectare-meters (ha·m)**.

- **Unit Trap:** If u is not in kg/ha·m/day, the formula will give the wrong answer.

14. Oxidation Ponds:

Large, shallow ponds that treat wastewater through the interaction of sunlight, bacteria, and algae (**Algae-Bacterial Symbiosis**). The algae produce oxygen via photosynthesis, which the bacteria use to break down the organics.

15. Septic Tanks:

A localized, **anaerobic** treatment unit typically used for individual households not connected to a main sewer line.

SECTION 7: THE 100% EXHAUSTIVE W9 Q&A BANK

Sewerage Systems & Wastewater Characteristics

- **Q:** Critical operational disadvantage of a Combined System of Sewerage?
 - **A:** It unnecessarily increases the hydraulic and organic load on the treatment plant, potentially leading to bypassing untreated sewage during storm events.
- **Q:** Correct hierarchy of sewer pipes from a house to the river? (Slide Veto)
 - **A:** House sewer Lateral sewer Trunk/main sewer Outfall sewer.
- **Q:** A wastewater sample reveals solids with particle size between $1 \mu\text{m}$ and $10^{-3} \mu\text{m}$. How are these classified?
 - **A:** Colloidal solids; they do not settle readily and require coagulation or biological treatment.
- **Q:** Average standard BOD₅ of domestic sewage per person is about:
 - **A:** 0.08 kg/day/person.
- **Q:** What is the concept of 'Population Equivalent' used for?

- **A:** To express the strength of industrial wastewater in terms of the equivalent number of people producing that much BOD.

Dissolved Oxygen (DO) & BOD Kinetics

- **Q:** How does DO saturation change with temperature?
 - **A:** DO saturation decreases as temperature increases (thermal pollution can worsen oxygen deficiencies).
- **Q:** Which holds more dissolved oxygen at the same temperature: freshwater or seawater? (Slide Veto)
 - **A:** Freshwater holds more dissolved oxygen.
- **Q:** Why is dilution of the wastewater sample required in a BOD test?
 - **A:** To ensure the oxygen demand does not exhaust the available DO, maintaining a residual DO of at least 1 mg/L.
- **Q:** What four nutrient solutions are added to the dilution water during a BOD test? (Slide Veto)
 - **A:** Calcium chloride, Magnesium sulphate, Phosphate buffer, and Ferric chloride.
- **Q:** How is Stage 1 distinct from Stage 2 in a standard BOD curve?
 - **A:** Stage 1 = Carbonaceous (CBOD) — heterotrophic bacteria oxidize organic carbon. Stage 2 = Nitrogenous (NBOD) — nitrification of ammonia.
- **Q:** How does increased temperature affect Ultimate BOD (L_0) and reaction rate constant (K)?
 - **A:** K increases, but L_0 remains constant.
- **Q:** Formula to adjust BOD reaction rate constant for different temperatures? (Slide Veto)
 - **A:** $K_T = K_{20} \times (1.047)^{(T-20)}$.
- **Q:** If BOD_5/COD ratio is between 0.63 and 0.68, what does this indicate? (Slide Veto)
 - **A:** The wastewater can be considered fully biodegradable.
- **Q:** If BOD_u/COD ratio is between 0.92 and 1.0, what does this indicate?
 - **A:** The waste is fully biodegradable.

Preliminary & Primary Treatment

- **Q:** Which treatment technique isn't a preliminary treatment of wastewater?
 - **A:** None of the above — Screening, Skimming, and Grit Chambers are ALL classified as preliminary treatment.
- **Q:** Which treatment technique is used to remove oil and grease from wastewater?
 - **A:** Skimming.
- **Q:** In a Grit Chamber, what happens if velocity drops significantly below 0.2 – 0.3 m/s?

- **A:** Lighter organic matter settles out along with the grit, leading to septic conditions and odors.
- **Q:** Which hydraulic device is installed in a grit chamber to keep flow velocity constant?
 - **A:** Proportional Weir (Sutro Weir).
- **Q:** Why are Parshall Flumes often preferred over weirs in inlet channels?
 - **A:** They are self-cleaning and cause less head loss compared to weirs.
- **Q:** Which type of settling is characteristic of Primary Sedimentation Tanks?
 - **A:** Type II: Flocculent Settling.

Secondary (Biological) Treatment

- **Q:** Which of the following is an anaerobic treatment unit?
 - **A:** Septic Tank.
- **Q:** Oxidation Ponds work on the principle of:
 - **A:** Algae-bacterial symbiosis.
- **Q:** Primary mechanism of pollutant removal in a Trickling Filter?
 - **A:** Adsorption of organics onto a microbial slime layer attached to the media, followed by aerobic oxidation.
- **Q:** Which problems are commonly encountered in a trickling filter?
 - **A:** All of the above — Ponding, Psychoda/filter flies, and Odor problems.
- **Q:** What is 'ponding' in a trickling filter, and how is it remediated?
 - **A:** Clogging of filter voids by excessive growth of algae/fungi; remediated by adding Chlorine or Copper Sulphate.
- **Q:** Which is NOT a primary benefit of recirculation in trickling filters?
 - **A:** "It reduces the hydraulic load on the filter." — This is FALSE. Recirculation actually **increases** the hydraulic load.
- **Q:** What is the NRC formula for calculating the efficiency (η) of a trickling filter? (Slide Veto)
 - **A:** $\eta = 100 / (1 + 0.0044 \times \sqrt{u})$, where u is the organic loading in **kg/ha·m/day**.

Numerical Calculations

- **Q:** BOD test: initial DO of 2% diluted sample = 6 mg/L; final DO after 5 days = 2 mg/L. Find BOD.
 - **A:** 200 mg/L. (BOD = $(6 - 2) / 0.02 = 200$ mg/L).
- **Q:** Volume of sample = 6 mL; dilution water = 294 mL; initial DO = 7.5 mg/L; final DO = 3.5 mg/L. Calculate BOD₅.
 - **A:** 200 mg/L. ($P = 6/300 = 0.02$; BOD = $(7.5 - 3.5) / 0.02 = 200$ mg/L).
- **Q:** If $K_{20} = 0.23 \text{ d}^{-1}$, find K_{15} and K_{30} . (Slide Veto)

- **A:** $K_{15} = 0.182 \text{ d}^{-1}$ and $K_{30} = 0.364 \text{ d}^{-1}$. (Using $K_T = 0.23 \times (1.047)^{(T-20)}$).
 - **Q:** Activated sludge process: 1 MLD flow, aeration tank volume = 200 m³. Find hydraulic retention time.
 - **A:** 4.8 hours. ($\text{HRT} = 200 / 1000 \text{ m}^3/\text{day} = 0.2 \text{ days} \times 24 = 4.8 \text{ hours}$).
 - **Q:** Moisture content of sludge reduced from 98% to 93%. Find % decrease in volume.
 - **A:** ~71.5% (rounds to 75% in standard answer key). ($V_1(100 - 98) = V_2(100 - 93)$
 $2V_1 = 7V_2 \quad V_2 \approx 28.5\%$; Decrease = $100 - 28.5 = 71.5\%$).
 - **Q: A wastewater sample has a 5-day BOD of 200 mg/L at 20°C. If the reaction rate constant k (base 10) is 0.1/day, what is the Ultimate BOD (L_0)?**
 - **A: 292.4 mg/L.** (Math:
 $L_0 = y_5 / (1 - 10^{-0.1 \times 5}) \rightarrow 200 / (1 - 10^{-0.5}) \rightarrow 200 / 0.6838 \approx 292.4$).
 - **Q: Using the same sample, what would be the 10-day BOD (y_{10})?**
 - **A: 263 mg/L.** (Math:
 $y_{10} = 292.4 \times (1 - 10^{-0.1 \times 10}) \rightarrow 292.4 \times (1 - 0.1) \rightarrow 292.4 \times 0.9 = 263.16$).
 - **Q: A BOD test is conducted at 30°C. If $K_{20} = 0.23 \text{ day}^{-1}$ (base e), find K_{30} .**
 - **A: 0.364 day⁻¹.** (Math: $K_{30} = 0.23 \times (1.047)^{30-20} \rightarrow 0.23 \times 1.58 \approx 0.364$).
-

SECTION 8: ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: If the velocity in a grit chamber drops significantly below the 0.2 – 0.3 m/s range, what goes wrong? What device fixes this?

Answer: If it drops too low, lighter organic matter (poop) settles out along with the heavy grit (sand), causing severe odor and septic conditions. A Proportional Weir (Sutro Weir) or Parshall Flume is used to keep the velocity constant.

Quiz 2: Your trickling filter is starting to “pond” (water is pooling on top of the rocks instead of trickling down). What is causing this, and how do you fix it?

Answer: The microbial slime layer (algae/fungi) has grown too thick and is clogging the voids between the rocks. You fix it by dumping Chlorine or Copper Sulphate to kill the excess growth.

Feynman Challenge: Explain the concept of “Population Equivalent” as if you were talking to a 10-year-old.

Example Answer: Imagine a normal person produces exactly 1 small bag of garbage a day (0.08 kg of BOD). Now imagine a giant factory that produces 1,000 bags of garbage a day. Instead of saying the factory makes “80 kg of BOD,” it’s easier to say

*the factory makes as much pollution as **1,000 people!** It translates confusing factory pollution into a simple number of humans so city planners know how big to build the wastewater treatment plant.*